



Roll No:

BPHARM
(SEM. III) THEORY EXAMINATION 2025-26
PHARMACEUTICAL ENGINEERING

TIME: 3 HRS**M.MARKS: 75**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt all questions in brief.**10 x 02 = 20**

a.	Define Reynolds number and its significance.
b.	State any two differences between laminar and turbulent flow.
c.	What is Venturi meter used for?
d.	Define fluid energy mill.
e.	What is sieve analysis?
f.	State Fourier's law of heat conduction.
g.	What is flash distillation?
h.	Define equilibrium moisture content.
i.	What is filter aid? Give one example.
j.	Define super centrifuge.

SECTION B

2. Attempt any two of the following:**02 x 10 = 20**

a.	Explain the principle, construction and working of Venturi meter. Add its pharmaceutical applications.
b.	Describe size separation techniques. Explain cyclone separator with neat labelled diagram.
c.	Explain evaporation process. Discuss working, merits and demerits of climbing film evaporator.

SECTION C

3. Attempt any five of the following:**05 x 07 = 35**

a.	Explain Bernoulli's theorem and discuss its pharmaceutical applications.
b.	Discuss mechanism of size reduction. Explain fluid energy mill with advantages and limitations.
c.	Write a note on heat transfer by radiation and convection with suitable examples.
d.	Explain rate of drying curve and its significance in pharmaceutical drying.
e.	Describe construction, working and applications of spray dryer.
f.	Explain filtration theories and factors affecting filtration rate.
g.	Discuss corrosion in pharmaceutical plants. Explain any two methods of prevention.